

CTF Challenge: The Ghost Device

Category: Networking

Points: 150

Scenario

You're a junior network administrator at a secure facility. Recently, monitoring tools have recorded short bursts of ARP noise that briefly knock over local host discovery and confuse DHCP assignments. The senior admin captured a snapshot of the core router's ARP table during one of these micro-glitches before leaving — and left a note: *"All official hardware is from Cisco, Dell, HP, Intel, Apple, or Google. The ghost device isn't. Find the one that doesn't belong."*

This time the senior admin didn't label manufacturers in the table — he expects you to do the OUI lookups yourself and also to recognize MAC-level anomalies that indicate spoofing or custom devices.

Your task: analyze the ARP table, identify the IP address of the device from a **non-standard** / suspicious manufacturer (the ghost), and generate the flag.

ARP Table Snapshot

(Fields shown: Protocol, Address, Age (min), Hardware Addr, Manufacturer — **manufacturer column intentionally blank; do OUI lookups**)

Protocol	Address	Age (min)	Hardware Addr	Manufacturer
Internet	192.168.1.1	-	5C-F9-DD-7B-1A-01	-
Internet	192.168.1.12	45	00-0A-95-8C-3E-F2	-
Internet	192.168.1.34	72	00-1A-2B-3C-4D-5E	-
Internet	192.168.1.46	3	1C-B7-2C-8A-F3-E6	-
Internet	192.168.1.53	0	02-1A-2B-7C-9D-0E	-
Internet	192.168.1.68	19	F8-E9-03-D3-A4-B5	-
Internet	192.168.1.79	7	AC-FD-EC-3A-B6-C7	-
Internet	192.168.1.88	15	00-1C-C0-55-66-77	-
Internet	192.168.1.94	2	D4-9A-20-A7-B8-C9	-

Internet	192.168.1.101	21	00-0C-29-BB-CC-DD	—
Internet	192.168.1.115	9	3C-D9-2B-DD-EE-FF	—
Internet	192.168.1.121	36	1A-2B-3C-4D-5E-6F	—
Internet	192.168.1.127	0	DB-44-58-A4-00-CF	—
Internet	192.168.1.132	4	B8-27-EB-7D-8E-9F	—
Internet	192.168.1.148	11	00-1B-63-C1-B2-A3	—
Internet	192.168.1.159	6	7C-AB-00-11-22-33	—

Notes for contestants:

- **Manufacturer** column is intentionally empty — you must perform OUI (first three octets) lookups to identify vendors.
- The network only uses devices from these approved vendors: **Cisco, Dell, HP (Hewlett Packard), Intel, Apple, Google**.
- The ghost device is *not* one of the approved vendors and demonstrates a subtle MAC-level anomaly.

What you must do / what counts as solving

1. For each MAC address, look up the OUI (first three octets) and determine the manufacturer.
2. Identify the entry whose MAC/OUI indicates it is **not** from an approved vendor *and* shows evidence of being a spoofed/locally-administered device (hint: inspect the first byte of the MAC).
3. Report the ghost device's **IP address** and build the flag.

Flag Format

The flag is a concatenation of the last digit of the ghost device's IP address plus its full MAC address (hex, uppercase, no hyphens or colons).

`l8xieee{LAST_DIGIT_OF_IP}{MAC_ADDRESS}`